1 Does contrastive decoding correct where the hallucination is, over the course of generation?

Context. Contrastive decoding (CD) tries to cut down hallucination by running a second, weaker branch on a corrupted image and subtracting its distribution from the expert's at every decoding step. In captioning this happens at every token, and the thing it is supposed to catch keeps moving as the sentence grows. So the natural question is a temporal one. As the caption gets longer, does CD keep pushing where the risk actually is? The short answer is no. In two lines of algebra and one plot we can see that the size of CD's push at each step is set by the KL between the two branches, that this KL falls off as the caption grows, and so CD pushes hardest at the very start, where there is almost nothing to fix, and has gone quiet by the time the late, image-dependent words show up.

The mechanism, in one connection. Say the expert sees the clean image x and the amateur sees the corrupted image x', both reading off the same generated prefix $y_{<t}$. Write the per-token gap between them as $\Delta_t(y) = \mathrm{logit}_A(y \mid x', y_{<t}) - \mathrm{logit}_E(y \mid x, y_{<t})$. CD's logit is $z^{CD} = (1+\alpha)z^E - \alpha z^A = z^E - \alpha\,\Delta_t$, so its correction at step t is just $-\alpha\,\Delta_t$. Now call the per-step KL between the branches the perturbation sensitivity, $\mathrm{PS}(t) = \mathrm{KL}[p_E(\cdot \mid x, y_{<t}) \,\|\, p_A(\cdot \mid x', y_{<t})]$. The amateur is just the expert with its logits tilted by the shift, $p_A \propto p_E\, e^{\Delta_t}$, and the KL of a tilt has a closed form (Appendix A), whose leading term is a variance:

$$\mathrm{PS}(t) = \psi_t(\Delta_t) - \mathbb{E}_{p_E}[\Delta_t] = \tfrac{1}{2}\,\mathrm{Var}_{p_E}[\Delta_t] + O(\Delta_t^3), \qquad \psi_t(\Delta) = \log \mathbb{E}_{p_E}\big[e^{\Delta}\big]. \tag{1}$$

So $\mathrm{PS}(t)$ is a squared (Fisher) measure of the perturbation size. CD's correction is the shift itself, with magnitude linear in Δ_t, so

$$\big\|\text{CD correction at } t\big\| = \alpha\|\Delta_t\| \;\propto\; \sqrt{\mathrm{PS}(t)}. \tag{2}$$

The KL curve and CD's correction schedule are therefore the same curve up to a square root. Turning α up or down only scales it, it never changes the shape.

Why the KL falls off, and why that is the wrong way round. Both branches use the same language weights. The only difference between them is the image. As the sentence gets longer, the words already written drive both branches' next-token logits in the same way and hold them together, so the one thing CD feeds on, their disagreement about the image, keeps shrinking toward a floor. Writing the shrinking shift as $\Delta_t \approx \Delta_0\, g(t)$ with g heading down, Eq. (1) gives $\mathrm{PS}(t) \approx \mathrm{PS}(0)\, g(t)^2$ and, by Eq. (2), a correction magnitude falling as $g(t)$. Hallucination risk does not behave like this. It follows how much the image still matters, and that fades much more slowly, since the picture stays relevant for the whole caption. So CD's push is loaded onto the front of the sentence while the risk sits at the back, and the two end up pulling in opposite directions in time. The picture to keep in mind: two people describe the same photo, one sees it sharp and one sees it blurred. On the first word they disagree a lot, because the blur is all that separates them. But once the sentence is rolling, both are mostly just finishing it off from the words already on the page, and the shared text pulls them back together. CD only acts where they disagree, so its push dies away right as the late, detail-heavy words, the ones most likely to be hallucinated, are being written.

Methodology. The whole argument leans on one empirical claim, that $\mathrm{PS}(t)$ falls with t, so we check it directly. For each prompt we run CD to get a caption, and at every generated position

t we grab the expert distribution $p_E(\cdot \mid x, y_{<t})$ and the amateur distribution $p_A(\cdot \mid x', y_{<t})$ over the same prefix, so both branches read exactly the same words and only the image input differs, and we take $\mathrm{PS}(t) = \mathrm{KL}[p_E \| p_A]$. We average this over prompts at matching positions on VisIT-Bench ($N = 578$) with LLaVA-1.5-7B, for three CD variants that differ only in how the amateur is built: VCD (a diffusion-noised image), SID (an information-degraded view), and ICD (a text-side instruction perturbation).

Results. All three variants tell the same story (Figure 1). The per-step KL climbs for a handful of tokens, then falls off monotonically and settles onto a positive floor, which is exactly the $\mathrm{PS}(t) = \mathrm{PS}(0)\, g(t)^2$ shape. VCD and SID each drop by about a factor of ten down the log axis. ICD sits a full order of magnitude lower and stays almost flat, so by Eq. (2) it is barely correcting anything at any step. In every case the early tokens is where KLD between amateur and expert is increasing(Exactly as predicted) and decreases as tokens increase. Read through Eq. (2), that is the temporal version of CD's miscalibration in generation: the push is biggest at the start of the caption, where the model is committing to ordinary syntactic, high-prior words at low risk, and has faded to its floor by the tail, where the late object and attribute words carry the image dependence that hallucination actually rides on.

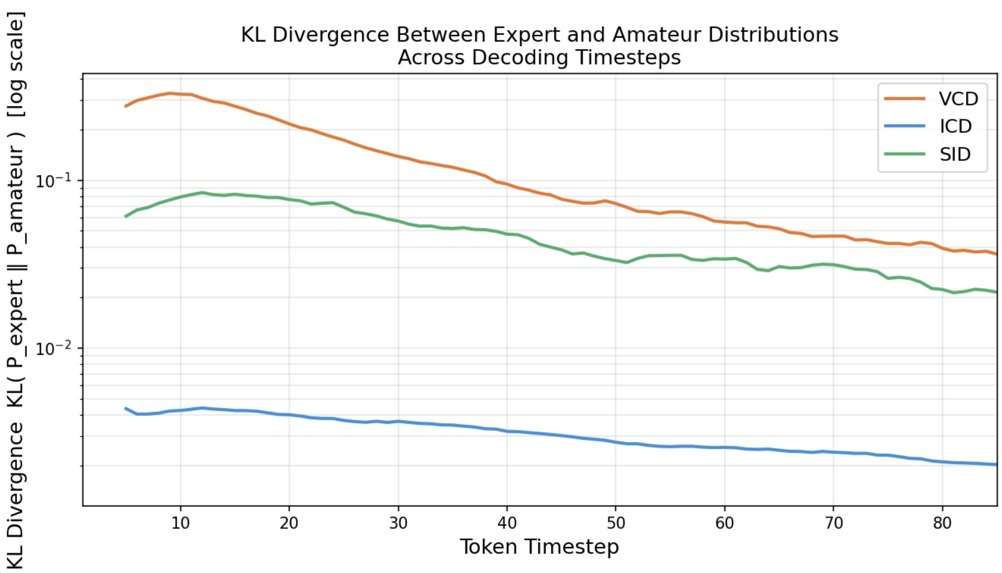

Figure 1: Per-step KL between expert and amateur, $\mathrm{PS}(t)$, against token position on VisIT-Bench ($N = 578$), log scale, for three CD variants. All fall off monotonically after a short early rise and level out on a positive floor. VCD and SID drop by roughly a factor of ten; ICD sits an order of magnitude lower and stays nearly flat. By Eq. (2) the correction size tracks the square root of these curves, so it too is spent on the early tokens and sitting near its floor by the tail, which is the opposite of where hallucination risk lives.

A The KL of a logit tilt

Since $z^A = z^E + \Delta_t$, the amateur is an exponential tilt of the expert,

$$p_A(y) = \frac{p_E(y)\, e^{\Delta_t(y)}}{\mathbb{E}_{p_E}[e^{\Delta_t}]}, \qquad \psi_t(\Delta) = \log \mathbb{E}_{p_E}\big[e^{\Delta}\big], \tag{3}$$

so the normalizer is the log-partition (cumulant generating) function ψ_t. The KL of a tilt is then exact,

$$\mathrm{PS}(t) = \mathrm{KL}[p_E \| p_A] = \psi_t(\Delta_t) - \mathbb{E}_{p_E}[\Delta_t] \ \geq 0 \tag{4}$$

(nonnegative by Jensen). Expanding the cumulant series $\psi_t(\Delta_t) = \mathbb{E}[\Delta_t] + \frac{1}{2}\,\mathrm{Var}[\Delta_t] + \cdots$, the mean cancels and the leading term is second order,

$$\mathrm{PS}(t) = \tfrac{1}{2}\,\mathrm{Var}_{p_E}[\Delta_t] + O(\Delta_t^3) = \tfrac{1}{2}\,\Delta_t^\top F_t\,\Delta_t + O(\Delta_t^3), \qquad F_t = \mathrm{Cov}_{p_E}, \tag{5}$$

a Fisher (squared) measure of the perturbation.

Since the KL is quadratic in Δ_t while CD's correction $-\alpha\Delta_t$ is linear, $\|\text{correction}\| \propto \sqrt{\mathrm{PS}(t)}$, as used in Eq. (2). With $\Delta_t \approx \Delta_0\, g(t)$ this gives $\mathrm{PS}(t) \approx \mathrm{PS}(0)\, g(t)^2$ and a correction magnitude falling as $g(t)$, so both decay monotonically with token position and both stay front-loaded relative to the slower-decaying vision relevance that hallucination tracks.